Calculating the probabilities for events having equally likely outcomes.

- A pregnancy yields a baby:

$$\Omega = \{ \text{boy}, \text{girl} \}$$

- Tossing a fair coin:

$$\Omega = \{ \text{head}, \text{tail} \}$$

- Rolling a 6-sided dice:

$$\Omega = \{ 1, 2, 3, \cdots, 6 \}$$

THM. If the Ω contains n equally likely w's, and each w has prob of $1/n$ being selected. Then $P(A) = \dfrac{|A|}{|\Omega|}$, where

$|A| = \#$ of w's in A.

Ex. Roll a 6-sided dice once. What is the prob of having the result < 3 ?

$\because$ $\Omega = \{1, 2, 3, \cdots, 6\}$

and $A = \{1, 2\}$

$\because$ $P(\{i\}) = \frac{1}{6}$

$\therefore$ $P(A) = \frac{|A|}{|\Omega|} = \frac{2}{6} = \frac{1}{3}$.

Review of counting principles

- The fundamental principle of counting

  - suppose a task
  - want to find out the # of ways to complete the task
  - separate the task into k stages
  - in each stage, there are $n_1, n_2, \ldots n_k$ # of ways to complete the sub-task.
  - the total # of ways to complete the original task $= n_1 \times n_2 \times \cdots \times n_k$

Ex. Toss 3 different 6-sided dices.

What is the # of ω's in Ω?

$$|\Omega| = \underset{1st}{\boxed{6}} \times \underset{2nd}{\boxed{6}} \times \underset{3rd}{\boxed{6}}$$

$$= 6^3$$

Ex. Place 3 different books on the shelf of a book case.

$$\#\text{ of placement} = \underset{1st}{\boxed{3}} \times \underset{2nd}{\boxed{2}} \times \underset{3rd}{\boxed{1}}$$

$$= 6$$

III Permutation : An $\boxed{\text{ordered}}$ arrangement of k objects out of n objects.

- When repetition ~~repetion~~ is allowed, then the # of ways to complete the task $= n^k$

(e.g. in the dice example, $n = 6$, $k=3$)

- When repetition is NOT allowed, then the # $= n! / (n-k)!$,

where $n! = n \cdot (n-1) \cdot (n-2) \cdot \cdots \cdot 2 \cdot 1$

(e.g. in the book case example, $n = 3$, $k=3$)

Combination : An unordered arrangement of k objects out of n objects.

☆ Repetition is NOT allowed in a combination problem.

$$\# \text{ of ways} = \binom{n}{k} = \frac{n!}{k!\,(n-k)!}.$$